

→ Backprop Intuition

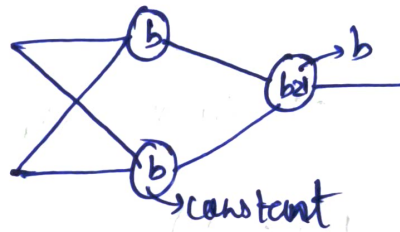
$$W_{\text{new}} = W_{\text{old}} - \eta \frac{\partial L}{\partial W}$$

$$\text{let } \eta = 1$$

$$W_{\text{new}} = W_{\text{old}} - \frac{\partial L}{\partial W}$$

$$L = (y - \hat{y})^2$$

$$L(b_{21}) \rightarrow \hat{y} = (b_{21})$$



a

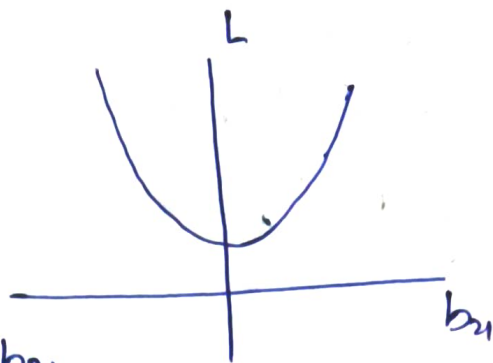
$$L = L(w, b) \rightarrow g \rightarrow \text{assume } g \text{ are constant}$$

~~b~~ ϕ_2

$L(b_{21}) \rightarrow \text{only variable}$

$$b_{21} = b_{21} - \frac{\partial L}{\partial b_{21}}$$

$$L = b_{21}^2$$



$\frac{\partial L}{\partial b_{21}} \rightarrow$ derivative of L w.r.t b_{21}

To reduce $L \downarrow$,

$\frac{\partial L}{\partial b_{21}} \Rightarrow +ve \rightarrow b_{21} \uparrow \quad L \uparrow$ (then we decrease b_{21} to decrease L)

$\frac{\partial L}{\partial b_{21}} = -ve \rightarrow b_{21} \downarrow \quad L \downarrow$ (then we increase b_{21} to reduce L)

Effect of learning rate (η)

It make the step ~~smooth~~ smooth.

What is Convergence

$$W_n = W_0 - \eta \frac{\partial L}{\partial W} = 0$$

$[W_n \approx W_0] \rightarrow$ It means we reach minima.

$\rightarrow$ What is vanishing gradient problem

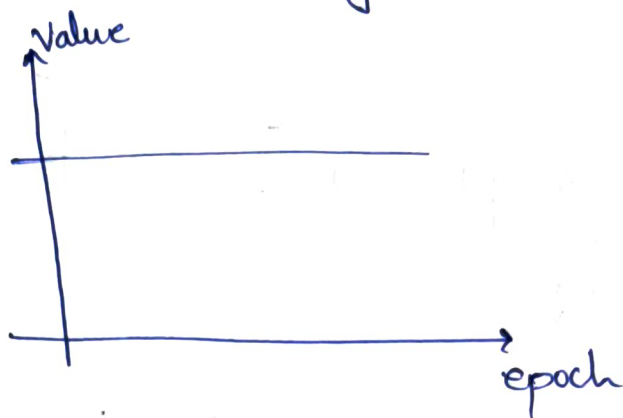
It occurs when training deep neural networks using backpropagation.

During backpropagation; gradients are passed from the output layers back to earlier layers.

If the activation function has derivatives less than 1 then multiplying many small derivatives together causes the gradient to become extremely small.

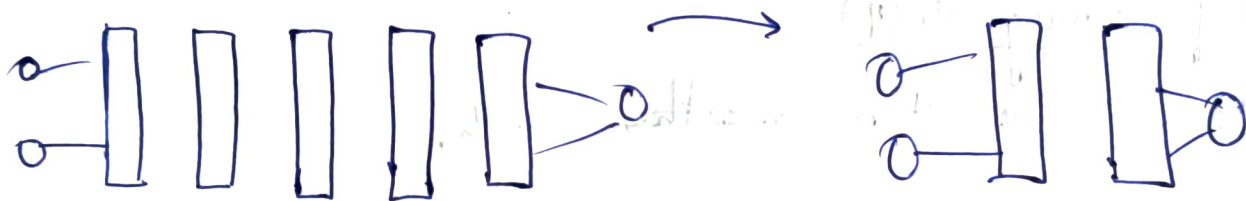
How to recognize:-

- ① Loss focus $\rightarrow$ If there is no change in loss, then it is VGP.
- ② Weight $\rightarrow$ graph.



$\rightarrow$ How to ~~Re~~ handle Vanishing Gradient Problem:-

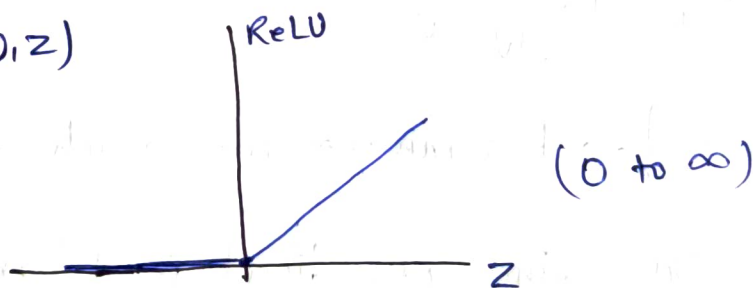
- ① Reduce Model Complexity.



- ② Using ReLU Activation functions.

$$\text{ReLU} = \max(0, z)$$

if $z > 0$, output = z
if $z \leq 0$, output = 0



$$\frac{d}{dz} \text{ReLU}(z) = 1$$

$$1 \times 1 \times 1 = 1$$

Problem $\rightarrow$ Dying ReLU

If a neuron always receives negative inputs $\text{ReLU}(z) = 0$, then that neuron stops learning permanently. This is called the Dying ReLU problem.

③ Proper weight initialization → Glorot
→ Xavier.

⑭

④ Batch Normalization → layers

⑤ Residual Network.

↳ building block → ANN